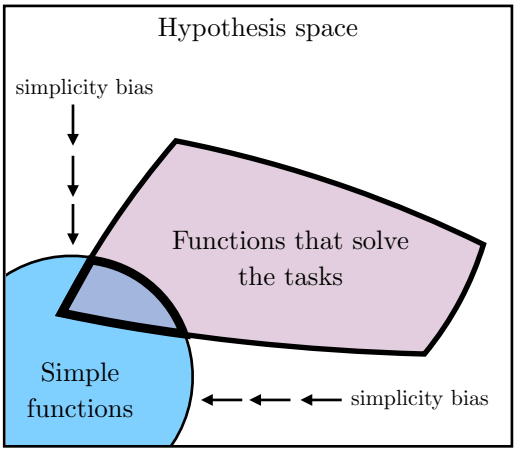

Figure 7. **The Simplicity Bias Hypothesis:** Larger models have larger coverage of all possible ways to fit the same data. However, the implicit simplicity biases of deep networks encourage larger models to find the simplest of these solutions.

The Capacity Hypothesis

Bigger models are more likely to converge to a shared representation than smaller models.

3.3. Convergence via Simplicity Bias

Arriving at the same mapping on the *training data* does not prohibit the models from developing distinct internal representations. It is not unreasonable to posit that the representations used to detect a dog in a 1M parameter model could be quite different than that used by a 1B parameter model. What would stop a billion-parameter (and counting) model from learning an overly complicated and distinct representation? One key factor might be simplicity bias:

The Simplicity Bias Hypothesis

Deep networks are biased toward finding simple fits to the data, and the bigger the model, the stronger the bias. Therefore, as models get bigger, we should expect convergence to a smaller solution space.

Such simplicity bias could be coming from explicit regularization $\mathcal{R}(f)$ commonly used in deep learning (*e.g.*, weight decay and dropout). However, even in the absence of external influences, deep networks naturally adhere to Occam's razor, implicitly favoring simple solutions that fit the data (Solomonoff, 1964; Gunasekar et al., 2018; Arora et al., 2019a; Valle-Perez et al., 2019; Huh et al., 2023; Dingle et al., 2018; Goldblum et al., 2023). Figure 7 visualizes how simplicity bias can drive convergence.

4. What representation are we converging to?

By now, we hope to have convinced the reader that task and data pressures, combined with increasing model capacity, can lead to convergence. We next turn our attention to *what* exactly is the endpoint of all this convergence.

Our central hypothesis, stated in Figure 1, is that the representation we are converging toward is a statistical model of the underlying reality that generates our observations. Consistent with the multitask scaling hypothesis, such a representation would naturally be useful toward many tasks (or at least toward any task grounded in reality). Additionally, this representation might be relatively simple, assuming that scientists are correct in suggesting that the fundamental laws of nature are indeed simple functions (Gell-Mann, 1995), in line with the simplicity bias hypothesis.

But what exactly do we mean by "a statistical model of the underlying reality." In this section, we formalize one definition with concrete mathematical statements. *Importantly*, this section should be read as just one concrete candidate for the form of the platonic representation; other candidates could be arrived at from other modeling assumptions.

4.1. An idealized world

We consider a world that works as follows, consistent with the cartoon in Figure 1. The world consists of a sequence of T discrete events, denoted as $\mathbf{Z} \triangleq [z_1, \ldots, z_T]$, sampled from some unknown distribution $\mathbb{P}(\mathbf{Z})$. Each event can be observed in various ways. An observation is a bijective, deterministic function $\texttt{obs} : \mathcal{Z} \to \cdot$ that maps events to an arbitrary measurement space, such as pixels, sounds, mass, force, torque, words, etc. Later, in Section 6, we discuss limitations and potential extensions to continuous and unbounded worlds, and stochastic observations, that could yield a model that better reflects real learning scenarios.

One can think of an event as corresponding to the state of the world at some point in time[3], but it is also fine to simply consider an event as any variable that indexes observations, with no further physical meaning[4].

In this idealized world, knowing $\mathbb{P}(\mathbf{Z})$ would be useful for many kinds of predictions; this would constitute a world model over the events that cause our observations (Werbos, 1987; Ha & Schmidhuber, 2018; Richens & Everitt, 2024). We will next show that a particular representation of $\mathbb{P}(\mathbf{Z})$ is recovered by certain contrastive learners.

[3]Here we only analyze temporal sequences, but note that the same could be done with respect to events laid out in space instead.

[4]This latter interpretation may be more consistent with Plato's intent. Scholars have argued that his allegory of the cave rejects any notion of a true world state (Nettleship, 1897). Instead, we could say that the joint distribution of observation indices is *itself* the platonic reality.

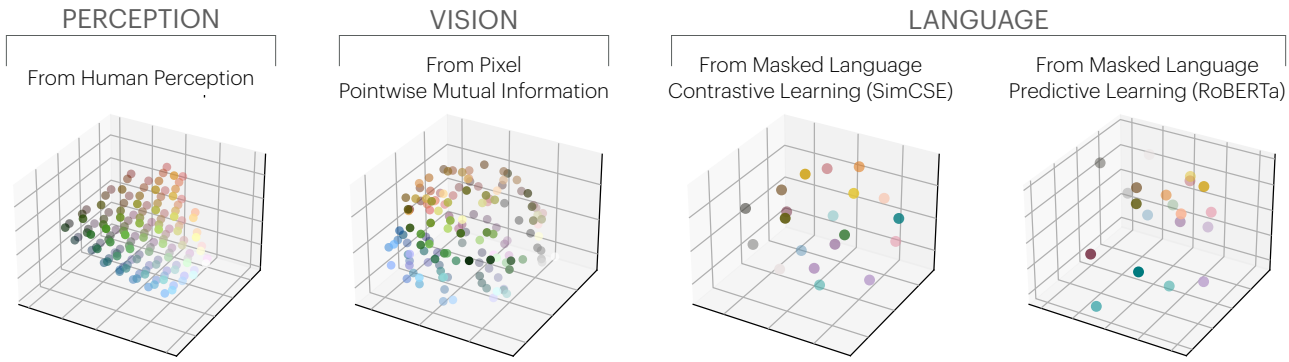

Figure 8. **Color cooccurrence in VISION and LANGUAGE yields perceptual organization:** Similar representations of color are obtained via, **from LEFT to RIGHT**, the perceptual layout from CIELAB color space, cooccurrence in CIFAR-10 images, and language cooccurrence modeling (Gao et al. (2021); Liu et al. (2019); computed roughly following Abdou et al. (2021)). Details in Appendix D.

4.2. A family of contrastive learners converge to a representation of $\mathbb{P}(\mathbf{Z})$

Consider a contrastive learner that models observations that *cooccur* together. For simplicity, we ground our discussion with the following definition of the *cooccurrence probability*, P_{coor}, of two observations x_a and x_b both occurring within some window T_{window}:

$$P_{\mathsf{coor}}(x_a, x_b) \quad \propto \sum_{(t,t')\colon |t-t'| \leq T_{\mathsf{window}}} \mathbb{P}(X_t = x_a, X_{t'} = x_b).$$

Analogously, we can define P_{coor} for $\mathbf{Z}$ and other observation modalities. Note that P_{coor} is symmetric.

Consider *positive pairs* as two observations nearby in time (sampled from P_{coor}) and *negative pairs* as observations drawn from any point in time (sampled independently from the marginal). Our contrastive learner tries to classify if a pair is positive or negative by learning a representation $f_X \colon X \to \mathbb{R}^d$ such that the dot-product kernel approximates the log odds ratio up to some offset:

$$\langle f_X(x_a), f_X(x_b) \rangle \approx \log \frac{\mathbb{P}(\mathsf{pos} \mid x_a, x_b)}{\mathbb{P}(\mathsf{neg} \mid x_a, x_b)} + \tilde{c}_X(x_a) \quad (3)$$

$$= \log \frac{P_{\mathsf{coor}}(x_a \mid x_b)}{P_{\mathsf{coor}}(x_a)} + c_X(x_a) \quad (4)$$

$$= K_{\mathsf{PMI}}(x_a, x_b) + c_X(x_a), \quad (5)$$

where K_{PMI} is the pointwise mutual information (PMI) kernel, and $c_X(x_a)$ is constant in x_b. We note that this is a common setting for self-supervised contrastive learners with NCE objectives (Gutmann & Hyvärinen, 2010; Oord et al., 2018), including SimCLR (Chen et al., 2020) and SimCSE (Gao et al., 2021). (See Oord et al. (2018) and Appendix F.1 for detailed derivations.)

Under mild conditions that the world is smooth enough (see Appendix F.2), a choice of f_X can exactly represent K_{PMI}:

$$\langle f_X(x_a), f_X(x_b) \rangle = K_{\mathsf{PMI}}(x_a, x_b) + c_X, \quad (6)$$

where we observed that $c_X(x_a)$ from Equation (5) must be a constant since both sides are symmetric.

Therefore, the contrastive learners we consider are minimized by a representation f_X whose kernel is K_{PMI} (up to a constant offset). With sufficient data and optimization, we will observe convergence to this point.

Thus we have convergence to a representation of the statistics of X, but what about Z? Recall that our idealized world consists of *bijective* observation functions, which, over discrete random variables, preserve probabilities. So we have:

$$P_{\mathsf{coor}}(x_a, x_b) = P_{\mathsf{coor}}(z_a, z_b)$$
$$K_{\mathsf{PMI}}(x_a, x_b) = K_{\mathsf{PMI}}(z_a, z_b),$$

where we use P_{coor} and K_{PMI} in a modality-agnostic way to emphasize that different modalities share the same these quantities.

All these arguments hold not just for X but also for Y (or any other bijective, discrete modality), implying:

$$K_{\mathsf{PMI}}(z_a, z_b) = \langle f_X(x_a), f_X(x_b) \rangle - c_X \quad (7)$$
$$= \langle f_Y(y_a), f_Y(y_b) \rangle - c_Y. \quad (8)$$

Therefore, for any modality in our idealized world, we observe representational convergence to the same kernel, which represents certain pairwise statistics of $\mathbb{P}(\mathbf{Z})$.

This analysis suggests that certain representation learning algorithms may boil down to a simple rule: *find an embedding in which similarity equals PMI*. We note that this idea is consistent with prior works that have used PMI as a similarity measure for clustering in vision and language (*e.g.*, Isola et al. (2014); Isola (2015); Isola et al. (2016); Chambers & Jurafsky (2008)).

A study in color We conduct a case study to verify that convergence does happen on real data. Abdou et al. (2021)

discovered that color distances in learned language representations, when trained to predict cooccurrences in *text* (Devlin et al., 2018), closely mirror human perception of these distances, which we reproduce in Figure 8 with both contrastive and predictive models. Interestingly, they noted an increasing similarity as models scale larger and become better at modeling *text* cooccurrences. In Figure 8, we also learn representations of color based on K_{PMI} from cooccurrences in *images*. Indeed, learning cooccurrence statistics in either domain recovers roughly the *same* perceptual representation. Details of this experiment are described in Appendix D.

We believe that our simple model encapsulates essential aspects of complex real-world systems, and offers a path toward understanding the representation that models are converging to—a unified model that is proficient across various domains and modalities, grounded in the statistical properties of the underlying world. Section 6 further elaborates some limitations.

5. What are the implications of convergence?

Scaling is sufficient, but not necessarily efficient Our arguments are roughly in line with the claim that "scale is all you need" to reach high levels of intelligence. We have argued that as resources are scaled (# parameters, # datapoints, # flops), representations are converging, regardless of other modeling choices and even data modality. Does this mean that scale is all that matters? Not quite: different methods can scale with different levels of *efficiency* (Hestness et al., 2017; Kaplan et al., 2020), and successful methods must still satisfy some general requirements (*e.g.*, be a consistent estimator, model pairwise statistics of $\mathbb{P}(\mathbf{Z})$).

Training data can be shared across modalities Suppose you have access to N images and M sentences, and want to learn the best representation. If there is indeed a modality-agnostic platonic representation, then *both* image and language data should help find it. The implication is that if you want to train the best vision model, you should train not just on N images but also on M sentences. This is already becoming common practice (OpenAI, 2023; Radford et al., 2021). Many vision models are finetuned from pre-trained LLMs. The other direction is less common, but also is implied by our hypothesis: if you want to build the best LLM, *you should also train on image data*. Indeed, OpenAI (2023) showed that training on images improved performance on text. In theory, there should be some conversion ratio: a pixel is worth a words for training LLMs, and a word is worth b pixels for training vision models.

Ease of translation and adaptation across modalities When two representations are aligned, transitioning from one to the other should be a simple function that's easily obtained. Our hypothesis could explain the phenomenon that conditional generation is easier than unconditional (Mirza & Osindero, 2014; Liu et al., 2020; Sauer et al., 2022), as the data we condition on may have the same platonic structure as the data we are generating. In line with this, recent work has found that representation-conditioning is even easier (Li et al., 2023). Similarly, representational convergence could act as a bridge that lets us find mappings between domains even without paired data; this may underlie the success of unpaired translation in vision (Zhu et al., 2017; Shi et al., 2024; Xie et al., 2022) and language (Tran et al., 2017; Lample et al., 2018). We emphasize that this doesn't mean that models trained on a single modality (*e.g.*, language) can immediately process raw data from another (*e.g.*, vision). What makes them adaptable to the new modalities is that they share a common modality-agnostic representation, and can readily process *representations* of new modalities. Furthermore, this implies that language models would achieve some notion of grounding in the visual domain even in the absence of cross-modal data[5]. The primary advantage of cross-modal data could then simply be sample efficiency.

Scaling may reduce hallucination and bias A prominent shortcoming of current LLMs is their propensity to hallucinate, or output false statements. If models are indeed converging toward an accurate model of reality, and scale powers this convergence, then we may expect hallucinations to decrease with scale. Of course, our hypothesis is conditioned on the training data for future models constituting a sufficiently lossless and diverse set of measurements. This may not come to pass, but it is an implication of our hypothesis worth pointing out. A similar argument can be made about certain kinds of bias. It has been shown that large models can exacerbate existing biases present in their training data (Hall et al., 2022). Our hypothesis implies that, while this may be true, we should expect *larger* models to amplify bias *less*. This does not mean bias will be removed, rather that the model's biases will more accurately reflect the data's biases, rather than exacerbating them.

6. Counterexamples and limitations

Different modalities may contain different information One immediate objection to our hypothesis is: what about the information that is unique to a given modality? Can language really describe the ineffable experience of watching

[5] In 1688, William Molyneux asked if a person born blind, upon gaining sight, could distinguish shapes by vision alone (Locke, 1690). Our arguments suggest they could not do so immediately, but after some visual experience, they could easily map shapes to their prior touch-based representations. Empirical data supports this, showing that congenitally blind children given sight can quickly learn these abilities (Held et al., 2011).